

Assignment: Functions and Conditional Execution in Python

Course: Introduction to Programming with Python

Level: BS (Undergraduate)

Duration: 40–50 Minutes

Marks: 20

Instructions

1. Attempt all questions.
2. Write clean and properly indented Python code.
3. Use functions wherever required.
4. Use conditional execution (**if**, **elif**, **else**) appropriately.
5. Do not use AI-generated or copied code without understanding it.

Question 1 — Basic Function (4 Marks)

Create a function named `check_even_odd()` that:

- takes one integer as input,
- returns "Even" if the number is even,
- otherwise returns "Odd".

Sample Input

7

Expected Output

Odd

Question 2 — Grade Calculator Using Functions (6 Marks)

Write a function named `calculate_grade()` that:

- accepts marks (0–100),
- returns grade according to the following conditions:

Marks	Grade
80 and above	A
70–79	B
60–69	C
50–59	D
Below 50	Fail

Requirements

- Use `if-elif-else`.
- Call the function using user input.
- Display the returned grade.

Question 3 — Temperature Decision System (5 Marks)

Create a function named `weather_advice()` that:

- accepts temperature as input,
- gives advice according to the following rules:

Temperature	Message
Above 35	Stay hydrated
25–35	Weather is pleasant
Below 25	It may feel cold

Example

Input: 38

Output: Stay hydrated

Question 4 — Debugging Question (5 Marks)

The following program contains logical and syntax mistakes. Identify and correct the errors.

```
def compare(a,b)
    if a > b:
        print("A is greater")
    else
        print("B is greater")
```

```
compare(10,5)
```

Requirements

1. Rewrite the corrected code.
2. Briefly explain the mistakes.

Bonus Challenge (Optional)

Create a function named `login_system()` that:

- asks for username and password,
- prints "Login Successful" if:

- username = "admin"
 - password = "python123"
- otherwise prints "Invalid Credentials".

Submission Guidelines

- Submit a .py file.
- File name format:

YourName_Functions_Assignment.py

- Late submissions may receive deduction in marks.